STATEMENT OF WORK

Electrical Upgrades to the Grant County Aerial Tanker Base & Fire Cache
Gila National Forest, New Mexico

Construction Specifications

1. PROJECT OVERVIEW

- This project consists of furnishing and installing a complete, code-compliant electrical distribution system to support a designated concrete pad area at the Grant County Aerial Fire Tanker Base and Fire Cache, and the upgrade of an existing electrical/motor control panel serving the retardant pumping system

The work includes:

- The upgrade of an existing electrical/motor control panel serving the retardant pumping system
- Extension of electrical service to a designated concrete pad area

The completed system shall be suitable for continuous outdoor service in a high-use industrial environment and designed for durability, safety, and future expansion.

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2. APPLICABLE CODES AND STANDARDS

All work shall comply with:

- NFPA 70 (National Electrical Code – NEC)
- NFPA 70E (Electrical Safety)
- NEMA standards
- UL listing requirements
- State of New Mexico electrical codes
- Local Authority Having Jurisdiction (AHJ) requirements
- USDA Forest Service engineering standards (as applicable)

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3. GENERAL REQUIREMENTS

The Contractor shall:

- Furnish all labor, materials, equipment, and supervision
- Perform a **site verification walkdown** prior to construction
- Verify all existing electrical conditions and available capacity
- Coordinate all work with the Contracting Officer's Representative (COR)

Installation Quality

- All materials shall be **new, UL-listed, and rated for outdoor use**
- All installations shall be **neat, workmanlike, and mechanically secure**

Trenching & Surface Restoration

- Minimize trenching where feasible by utilizing existing infrastructure
- Restore all disturbed asphalt, concrete, or native surfaces to original condition

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4. ELECTRICAL DISTRIBUTION SYSTEM

4.1 Source of Power

All new outlets shall be served from a new service pole, location subject to COR approval.

4.2 Load Design

The Contractor shall:

- Perform NEC-compliant load calculations (Article 220)
- Design for:
    - Existing loads
    - New receptacles
    - Minimum **25% spare capacity**

4.3 Voltage Drop

- Maximum:

- **3% per branch circuit**
- **5% total system**

Conductor sizes shall be increased as required.

4.4 Wiring Methods

- Above ground: **EMT or Rigid Metal Conduit (RMC)**
- Underground:
  - **Schedule 40 PVC (minimum)**
  - **Schedule 80 PVC at risers**
- Install pull strings in all conduits

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5. RETARDANT PUMP ELECTRICAL EQUIPMENT MODERNIZATION

5.1 Scope and Intent

The Contractor shall modernize the existing Square D Well-Guard Class 8940 assembly to provide a safe, reliable, and code-compliant electrical service disconnect and motor control system for the retardant pumping equipment.

The intent is to upgrade the existing system to current industry standards, eliminating obsolete or degraded components while maintaining or improving functionality.

PANEL UPGRADE PHOTOS in appendix for reference

5.2 Minimum Performance Requirements

The completed installation shall:

- **Provide a single, clearly identifiable disconnecting means**
- **Provide overcurrent protection and motor overload protection appropriate for the connected pump motor**
- **Maintain full operational capability of the existing pump system**
- **Be suitable for outdoor installation (NEMA 3R minimum)**
- **Be constructed of new, current production, UL-listed equipment**

5.3 Required System Configuration

The modernized system shall include, at a minimum:

- **Circuit breaker-based disconnecting means (in place of or equivalent to existing fused disconnect, where appropriate)**
- **Motor controller (contactor or equivalent) for operation of the pump**
- **Motor overload protection in accordance with NEC Article 430**
- **Properly rated enclosure, conductors, and terminations**

Critical Requirement

Replacement of the existing equipment with a standard distribution panelboard alone is not acceptable.

The system shall include dedicated motor control and overload protection suitable for the pump application.

5.4 Code Compliance

All work shall comply with:

- **NFPA 70 (NEC), including:**
  - **Article 230 – Services**
  - **Article 240 – Overcurrent Protection**
  - **Article 250 – Grounding and Bonding**
  - **Article 430 – Motors and Motor Circuits**
- **All applicable UL, NEMA, and AHJ requirements**

5.5 Modernization Requirements

The Contractor shall:

- **Remove or replace obsolete, damaged, or non-compliant components, including:**
  - **Fuse holders (if no longer appropriate)**

  - **Contactors and relays**
  - **Control wiring and terminations**
- **Install equipment that reflects current industry standards for safety and maintainability**
- **Ensure all equipment is:**
  - **Properly rated for voltage, phase, and available fault current**
  - **Compatible with the existing electrical system**

5.6 Grounding and Bonding

The Contractor shall:

- **Inspect and upgrade grounding and bonding systems as required**
- **Ensure full compliance with NEC Article 250**
- **Verify continuity of all grounding conductors**

5.7 Contractor Design Responsibility

The Contractor shall:

- **Verify:**
  - **System voltage and phase**
  - **Motor nameplate data**
  - **Available fault current (as required)**
- **Replace existing panel with a complete, functional, new system**

5.8 Submittals

Prior to installation, the Contractor shall provide:

- **Equipment data sheets**
- **Confirmation of compliance with applicable NEC requirements**

5.9 Acceptance

Work will be accepted upon:

- **Verification of safe and proper operation of the pump system**
- **Inspection confirming compliance with all applicable codes**
- **Approval by the COR**

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6. RECEPTACLE OUTLET INSTALLATION (GENERAL)

6.1 Electrical Requirements

All receptacles shall be:

- **120V, 20A, single-phase**
- **GFCI protected (NEC 210.8)**
- Commercial/industrial grade

6.2 Enclosures

- Weatherproof: **NEMA 3R minimum**
- Covers:
    - **Extra-duty, in-use (bubble covers)**

6.3 Mounting

- Height: **42”–48” AFF**
- Mounted to structural members/posts using corrosion-resistant hardware
- Rigid and secure installation required

6.4 Orientation

- Face primary working area
- Dual-use posts shall have **back-to-back or dual-gang configurations**

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7. CONCRETE PAD ELECTRICAL SERVICE

Provide electrical service to the designated concrete pad site.

7.1 Scope

Install a branch circuit or feeder to supply receptacles at the future pad site.

7.2 Receptacle Requirements

Install:

- **Two (2) duplex receptacles total**
    - **One (1) on the east side**
    - **One (1) on the west side**

7.3 Electrical Specifications

Each receptacle shall:

- Be **120V, 20A**
- Be **GFCI protected**
- Be commercial/industrial grade

7.4 Mounting

- Mounted on:
    - Rigid posts, bollards, or approved supports
- Height:
    - **42”–48” AFF**
- Oriented toward working area

7.5 Enclosures

- Weatherproof: **NEMA 3R minimum**
- Covers:
    - **Extra-duty in-use covers**

7.6 Disconnect

- Provide **lockable, weatherproof disconnect switch** within sight of pad

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7.7 Conduit Installation

- Underground:
    - Schedule 40 PVC (Schedule 80 at risers)
- Above ground:
    - EMT or RMC
- Install expansion fittings where required

7.8 Future Capacity

- Provide **one (1) spare empty conduit** stubbed to pad
- System shall support future equipment loads

7.9 Voltage Drop

- Maximum **3% at receptacles**

7.10 Grounding

- Bond all metallic components
- Provide equipment grounding conductor
- Ensure continuity to panel

7.11 Distance and Routing Considerations

Electrical service to the concrete pad shall be routed from the new service pole, which shall be installed as part of the receptacle/stub installation.

The Contractor is advised that:

- The existing utility source is NOT located 30–50 feet from the pad. A new service pole IS REQUIRED to establish a new electrical source location. Once installed, the routed distance from the service pole to each receptacle will be approximately 30–50 feet.
- Routing distance to the west-side receptacle will be approximately 30–50 feet AFTER installation of the required service pole. The existing utility infrastructure does not currently support these distances without the pole.

Design Requirements

The Contractor shall:

- Verify actual field distances prior to installation
- Size all conductors to maintain:
    - **Maximum 3% voltage drop at each receptacle**
    - **Maximum 5% total system voltage drop**
- Increase conductor size as required to meet these limits (no exceptions)

System Configuration Responsibility

The Contractor shall be responsible for determining the most appropriate installation method, which may include:

- Direct branch circuit extension from an existing panel
- Installation of a subpanel (if justified by load or efficiency)
- Alternative routing methods to minimize distance and voltage drop

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New Service / Pole Installation

A new service pole SHALL be installed as part of this project. Its installation is REQUIRED to meet voltage drop and routing requirements and is included in the base scope of work.

A new service pole is required. The Contractor shall:

1. Submit:
    - Load calculations
    - Voltage drop calculations
    - Documentation confirming installation of the required service pole
2. Obtain Contracting Officer approval **prior to proceeding**

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8. IDENTIFICATION & LABELING

- Label all panels, disconnects, and circuits
- Use engraved or machine-printed labels
- No handwritten labeling permitted

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9. SUBMITTALS

Contractor shall provide all submittals and receive approval on all submittals prior to being issued a Notice to Proceed. No work can begin until the Notice to Proceed is issued from the Contracting Officer.

List of Submittals:

- Load calculations
- Panel schedule
- Conductor and conduit sizing
- Product data sheets
- As-built drawings

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10. TESTING & COMMISSIONING

Contractor shall:

- Verify voltage at all receptacles
- Test GFCI functionality
- Confirm grounding continuity
- Torque all terminations to manufacturer specs

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11. ACCEPTANCE

Work will be accepted upon:

- COR inspection and approval

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12. APPENDIX

Two (2) photos of the existing panel

Two (2) photos of the existing pump

One (1) Overhead map photo of approximate location for electrical outlets

Existing Panel in need of replacement:




L1
L2
L3
CLASS 8940 WELLGUARD CONTROLLER-TYPE 3R ENCLOSURE
TYPE
LINE VOLTS
MAX. HORSE POWER
OVER LOAD RELAY
REPLACEMENT PARTS
NEMA SIZE 4

Pump Photos:


GORMAN-RUPP
PUMPS
O SERIES
SELF-PRIMING
CENTRIFUGAL PUMP
MODEL NO.
04A20-B
IMP. DIA.
STD
SERIAL NO.
1730632
Gorman-Rupp Pumps
Mansfield, Ohio
GRPUMPS.COM
READ THE MANUALS FOR THIS PUMP THOROUGHLY BEFORE ATTEMPTING TO
INSTALL, OPERATE OR SERVICE THE PUMP.



Future Concrete Pad area and electrical outlets placement location - aerial photo (green ovals are approximately where electrical outlets shall be placed)

